

# Two-Part Semi-Continuous SEM: A Practical Guide

*For zero-inflated continuous outcomes (NSSI frequency, screen time, dissociation, etc.)*

## The problem this template solves

Many variables of clinical or behavioral interest are **zero-inflated**: a meaningful share of respondents score exactly zero (no symptoms, no episodes, no behavior), while everyone above zero forms a continuous, usually right-skewed, distribution. Non-suicidal self-injury (NSSI) frequency, problem-gambling symptom counts, disordered screen-time overuse, and dissociation-episode counts all share this shape.

Fitting a single linear or ordinary Poisson/negative-binomial model to such a variable conflates two conceptually distinct questions:

1. **Does the behavior occur at all?** (a threshold / onset process)
2. **How severe is it, once present?** (a continuous, graded process)

These two processes plausibly have different predictors, different effect sizes, and sometimes even different signs. A model that doesn't separate them will misrepresent both.

## The two-part (hurdle) solution

This template splits the outcome into:

- **Part A — Entry.** A probit latent-response SEM (WLSMV estimator) modeling the binary transition from “no symptoms” ( $Y = 0$ ) to “any symptoms” ( $Y > 0$ ). WLSMV/probit treats the binary indicator as a threshold on an underlying continuous propensity (Muthén, 1984), which is more defensible than treating a 0/1 split as an arbitrary logistic outcome.
- **Part B — Severity.** A continuous SEM (MLR estimator, robust to non-normality) modeling log-transformed severity **among those with  $Y > 0$  only**.

Both parts share the same mediation structure ( $X \rightarrow \text{mediator(s)} \rightarrow Y$ ), and both allow one or more continuous moderators on every mediator  $\rightarrow Y$  path, estimated via mean-centered interaction terms.

## Why this beats an arbitrary median split or a plain zero-inflated GLM

- A median split or clinical cutoff throws away information and is sensitive to where exactly the line is drawn. The two-part model keeps the full continuous severity information above the hurdle.
- A single zero-inflated Poisson/negative-binomial model is a reasonable alternative and is included here as **Section 6, a robustness check** — but it assumes a specific count-generating process and does not fit naturally within a latent-variable SEM framework with mediators and moderators the way the two-part approach does. Reporting both, and showing they agree, is what reviewers increasingly expect for zero-inflated outcomes.
- Reporting *only* one model invites a reviewer question (“did you check whether this holds under a different distributional assumption?”). This template answers that question pre-emptively.

## What's included

| File                                | Purpose                                                                                                                                                   |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>template_twopart_sem.R</code> | The full, generalized script. Edit the <b>CONFIGURATION</b> block at the top only — variable names for X, mediators, the zero-inflated Y, and moderators. |
| <code>toy_data.csv</code>           | 10 rows of illustrative data using the default configuration names, so you can run the script once, end-to-end, before pointing it at real data.          |

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## How to adapt it to a new outcome

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1. Identify your **X** (distal predictor), **M** (one or more mediators), your **zero-inflated Y**, and your continuous **W** (moderator(s)).
2. Edit the configuration block at the top of the script — nothing else needs to change for a straightforward 1–3 mediator, 1–2 moderator design.
3. Run the script against `toy_data.csv` first. Confirm it prints structural paths and (with real sample sizes) converges.
4. Point `data_path` at your real dataset and re-run.
5. Read the console output for: structural path table, conditional indirect effects at representative moderator values, indices of moderated mediation, and Johnson–Neyman boundaries. Figures are written to `fig_dir`.

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## What this template does *not* decide for you

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- Whether a probit/WLSMV vs. logistic/ML approach is more appropriate for your entry outcome (this is a modeling choice to discuss with a statistician or your advisor for unusual sample sizes or missingness patterns).
- Whether your specific mediators and moderators are theoretically justified — that is a substantive, not a methodological, decision.
- Sample-size adequacy. SEM with WLSMV is data-hungry; as a rule of thumb, treat  $N < 150$  as too small for Part A, and check the proportion of  $Y > 0$  cases feeding Part B separately.

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## A note on attribution

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## Reference

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Muthén, B. (1984). A general structural equation model with dichotomous, ordered categorical, and continuous latent variable indicators. *Psychometrika*, 49(1), 115–132.

Bauer, D. J., & Curran, P. J. (2005). Probing interactions in fixed and multilevel regression: Inferential and graphical techniques. *Multivariate Behavioral Research*, 40(3), 373–400.